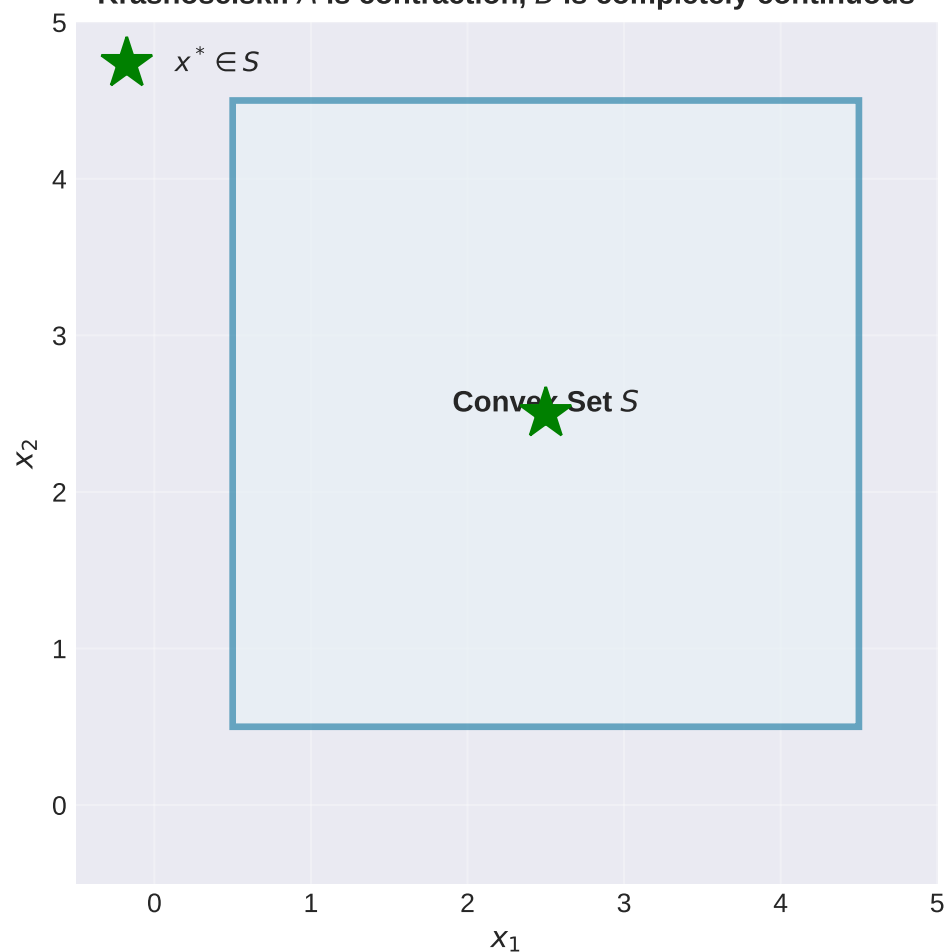


## Krasnoselski: $A$ is contraction, $B$ is completely continuous



### $\mathcal{P}$ -Lipschitzian Mappings

Definition:  
 $T$  is  $\mathcal{P}$ -Lipschitzian if:

$$\|Tx - Ty\| \leq \phi(\|x - y\|)$$

where  $\phi$  is a continuous function  
 with  $\phi(r) < r$  for  $r > 0$

Key property:

- Generalizes contractions and nonexpansive mappings
- Enables analysis of iterative methods in infinite dimensions

## Burton's Modification of Krasnoselski

Krasnoselski (1964):

- $A$ : contraction on  $S$
- $B$ : completely continuous
- $Ax + By \in S$

Burton (1995) weakened:

- $A$ : still contraction
- $B$ : maps  $S$  to bounded sets
- Weaker covering condition

Both guarantee fixed points for  
 operator equation:  $Ax + Bx = x$

### $\mathcal{D}$ -Lipschitzian Mappings

Definition (Dhage 2003):  
 $T$  is  $\mathcal{D}$ -Lipschitzian if:

$$\|Tx - Ty\| \leq \phi(\|x - y\|)$$

where  $\phi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$   
 is continuous, nondecreasing,  
 with  $\phi(0) = 0$

Key difference:

- No requirement that  $\phi(r) < r$
- More general than  $\mathcal{P}$ -Lipschitzian
- Includes broader class of mappings